LEARNER GUIDE

Forming and solving a pair

Equations, Inequalities and Simultaneous Relationships · Lesson 9. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To find **two** unknowns you need **two** facts. A situation like 'two lunches and a drink cost £8; three lunches and two drinks cost £13' hides two prices, and gives two sentences — and each sentence becomes an **equation**. Name the unknowns (let x and y be the two prices), form the **pair**, solve it with substitution or elimination, then **interpret**: say what each unknown is worth and check it fits both facts. Once you know each value, you can price any combination, not just the two you were told.

What you are learning to notice

- Two unknowns need two facts — each fact becomes one equation.
- 'Two lunches' $\rightarrow 2x$; 'a drink' $\rightarrow 1y$; 'cost £8' $\rightarrow = 8$. It is the same translation as before, done twice.
- Solve the pair with the method that suits it (elimination or substitution).
- Interpret: state each value in context, and check both facts — then you can answer new questions.

Moving through it, one step at a time

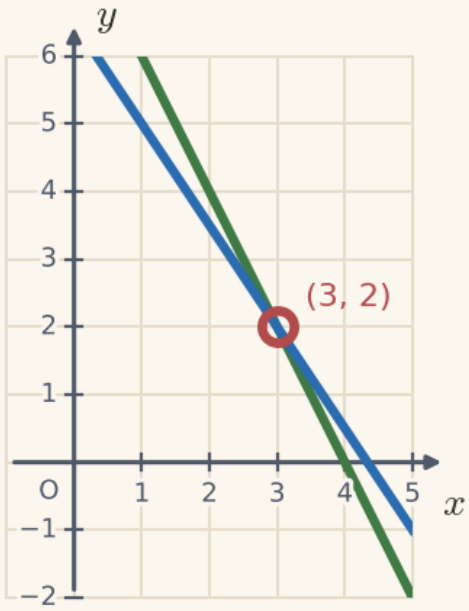
Name the two unknowns clearly. Read each sentence and turn it into an equation: the number of each item is a coefficient, the total is the right-hand side. Solve the pair — elimination is often quick when a variable matches, substitution when one is easily made the subject. State what each unknown is worth, check the pair in both original facts, and, if asked, use the values to cost a new combination.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A café order

Two lunches and a drink cost £8; three lunches and two drinks cost £13. Let x = lunch price, y = drink price.

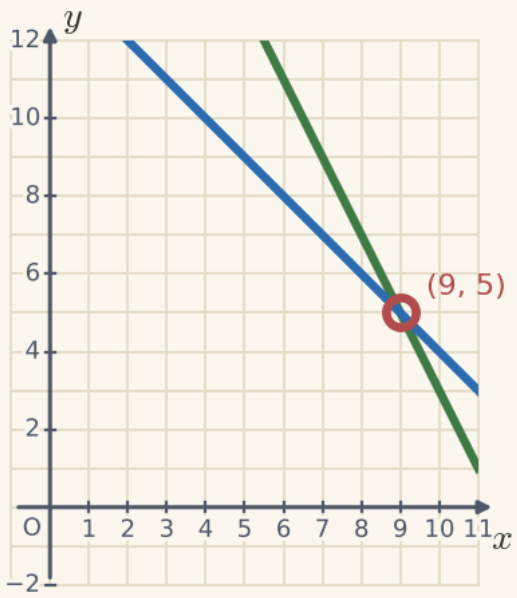


1. Form an equation from each sentence.	$2x + y = 8$ and $3x + 2y = 13$
2. Eliminate y: double the first, then subtract.	$(3x + 2y) - (4x + 2y) = 13 - 16 \rightarrow x = 3$
3. Find y and interpret.	$y = 2$ — a lunch is £3, a drink £2
4. Check both facts.	$2\times3 + 2 = 8$ ✓ and $3\times3 + 2\times2 = 13$ ✓

A lunch is £3 and a drink £2 — formed as $2x + y = 8$ and $3x + 2y = 13$, solved and checked in both facts.

Example 2 — Tickets

Two adult tickets and a child ticket cost £23; one adult and one child cost £14.



1. Form the pair (x = adult, y = child).	$2x + y = 23$ and $x + y = 14$
2. Subtract — the y matches.	$(2x + y) - (x + y) = 23 - 14 \rightarrow x = 9$
3. Find y and interpret.	$y = 5$ — an adult is £9, a child £5
4. Check both facts.	$2\times9 + 5 = 23$ ✓ and $9 + 5 = 14$ ✓

An adult ticket is £9 and a child £5 — subtracting removes y directly; both facts check.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (writing expressions from words, and solving a pair by substitution or elimination (Lessons 7–8)), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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